EXPERIMENTAL STUDY ON USAGE OF RECYCLED AGGREGATE IN CONCRETE IN BUILDING CONSTRUCTION

A Minor Project report Submitted in partial fulfilment of the requirement for the award of the Degree of

BACHELOR OF TECHNOLOGY

IN

CIVIL ENGINEERING

Submitted by

TUMMA HARSHA VARDHAN 19RJ5A0161

Under the Esteemed Supervision of

Mr.B.V. HARI PRASAD

Assistant professor

Department of Civil Engineering

MALLA REDDY INSTITUTE OF TECHNOLOGY

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MALLA REDDY INSTITUTE OFTECHNOLOGY

Department of Civil Engineering

CERTIFICATE

This is to certify that minor Project entitled **"EXPERIMENTAL STUDY ON USAGE OF RECYCLED AGGREGATE IN CONCRETE IN BUILDING CONSTRUCTION**" submitted by T.HARSHA VARDHAN (19RJ5A0161) in partial fulfilment for the award of the 'Bachelor of Technology in the Civil Engineering Department to Malla Reddy Institute of Technology, Maisammaguda, Secunderabad. Telangana State.

The results embodied in this minor Project have not been submitted to any other University or Institution for the award of any Degree or Diploma.

Approved and supervised by

Mr.B.V. HARI PRASAD
Assistant Professor
Department of Civil Engineering

EXTERNAL EXAMINER

Mr.ANITH KUMAR.T
Assistant Professor& Head of
Department of Civil Engineering

DECLARATION

We hereby declare that the contents presented in the Minor Project entitled **"EXPERIMENTAL STUDY ON USAGE OF RECYCLED AGGREGATE IN CONCRETE IN BUILDING CONSTRUCTION"** submitted in partial fulfilment for the award of B.Tech, degree to MALLA REDDY INSTITUTE OF TECHNOLOGY is a record of original work investigated by us under the supervision of **B.V.HARI PRASAD,** This is to state that the results embodied in this Minor Project have not been submitted to any other University or Institution for the award of any Diploma or Degree.

Submitted by

T. HARSHAVARDHAN 19RJ5A0161

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T. HARSHAVARDHAN 19RJ5A0161

ABSTRACT

This project mainly discus about the concrete production by mixing recycle aggregate to get the required strength for different applications by replacing the natural coarse aggregate. The Concrete which is made by replacing the recycled coarse aggregate strength can be determined by few physical and mechanical characteristics.

The collected recycle aggregate is used to make a concrete such that, several concrete specimens are prepared and evaluated with the virgin concrete. This project likewise talks about the examination of reuse aggregate with constant concrete blend of M25 and water cement proportion, similar to two unique ages have been taken and the age difference between those ages are of 30 years and few samples were caste and these two types of reuse aggregate is compared with conventional cement.

The 6 cubes were moulded with the conventional concrete with of the same M25 mix. The utilization of recycled coarse aggregate is the answer for the concrete waste and furthermore answers for the ecological issues and decreases the extraction of natural coarse aggregate which demonstrates the negative impact.

There is a lot of solid waste is generating day by day in world either it may in the form of construction waste or in the from industrial waste etc., that solid waste has to be used in the better way by recycling those material, in this project, I am going to deal with the construction solid waste which will come from the demolition of the concrete structure, from this concrete waste we can extract the aggregate and used for the exchange of the natural aggregate either it may be partly or completely and this type of extracted aggregate from the construction waste is called as recycled aggregate.

This recycled aggregate has been collected from the different places of different ages, it is well segregated, crushed, washed, sieved, and used for the experiment purpose with the mix proportion of M25, as a replacement of naturally available coarse aggregate in the proportions of 10%, 15% and 20%

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CHAPTER 1
INTRODUCTION

1.1 Problem Description

Urbanization growth rate in India is very high due to industrialization. Growth rate of India is reaching 9% of GDP. Rapid infrastructure development requires a large quantity of construction materials, land requirements & the site. For large construction, concrete is preferred as it has longer life, low maintenance cost & better performance. For achieving GDP rate, smaller structures are demolished & new towers are constructed. Protection of environment is a basic factor which is directly connected with the survival of the human race. Parameters like environmental consciousness, protection of natural resources, sustainable development, play an important role in modern requirements of construction works. Due to modernization, demolished materials are dumped on land & not used for any purpose. Such situations affect the fertility of land. As per report of Hindu online of March 2007, India generates 23.75 million tons demolition waste annually. As per report of Central Pollution Control Board (CPCB) Delhi, in India, 48million tons solid waste is produced out of which 14.5-million-ton waste is produced from the construction waste sector, out of which only 3% waste is used for embankment.

Out of the total construction demolition waste, 40% is of concrete, 30% ceramics, 5% plastics, 10% wood, 5%metal, & 10% other mixtures. As reported by global insight, growth in global construction sector predicts an increase in construction spending of 4800 billion US dollars in 2013. These figures indicate a tremendous growth in the construction sector, almost 1.5 times in 5 Years. For production of concrete, 70-75% aggregates are required. Out of this 60-67% is of coarse aggregate & 33-40% is of fine aggregate. As per recent research by the Fredonia group, it is forecast that the global demand for construction aggregates may exceed 26 billion tons by 2012.

Leading this demand is the maximum user China 25%, Europe 12% & USA 10%, India is also in top 10 users. From environmental point of view, for production of natural aggregates of 1 ton, emissions of 0.0046 million ton of carbon exist whereas for 1 ton recycled aggregate produced only 0.0024-million-ton carbon is produced. Considering the global consumption of 10 billion tons/year of aggregate for concrete production, the carbon footprint can be determined for the natural aggregate as well as for the recycled aggregate.

The use of recycled aggregate generally increases the drying shrinkage creep & porosity to water & decreases the compression strength of concrete compared to that of natural aggregate

concrete. It is nearly 10-30% as per replacement of aggregate. Recycling reduces the cost (LCC) by about 34-41% & CO2 emission (LCCO2) by about 23-28% for dumping at public / private disposal facilities.

1.2 Objective of the project

The experiment was carried out to overcome the problems created due to huge requirement of the raw material for manufacturing of conventional building material and also to minimize hazards caused by industrial waste on the environment, some other objectives are:

- To use the demolished and construction waste aggregate in the new concrete as the recycled concrete aggregate reduces the environmental pollution as well as providing an economic value for the waste material.
- To study the utilization of demolished and construction waste as a replacement of natural coarse aggregate.
- To study the physical properties of demolished and construction waste aggregate by conducting experimental work.
- To development of alternate low cost and environment suitable building materials from industrial wastes in an economical way.
- Importance must be given to cheap and locally available building materials and hence it is necessary to check and utilize the suitable waste products to replace some of the coarse aggregate.

Dismantling Of the Material

The concrete from the dismantled buildings is taken and by using hammer we break the concrete blocks into 20mm size aggregates as we do this manually, we get irregular shapes and hence after preparing of the recycled aggregate we need to sieve the material through IS 20mm sieve and hence we get the required size of the aggregate.

The following figures show the preparation of the recycled aggregate from the dismantled building material

Figure 1.1: Concrete waste

Figure 1.2: Recycled Coarse Aggregate

Outline of the Project

The methodology we follow for recycling of concrete are as follows

- Select a grade of concrete for which a conventional mix is to be done.
- Now, materials required for preparation for conventional concrete must be brought and material testing must be done in the laboratory before mix is prepared.
- All the physical tests on the aggregates have to be done according to

IS:2386 (PART I) - Particle size and shape

IS:2386 (PART III) - Specific gravity, Density, Voids, Absorption and Bulking

- Also find the properties of the cement which is used in the mix
- Now, using IS 10262:2009 design the suitable mix for required grade of a conventional concrete by considering the physical properties of the material.
- Using the above mix design prepare the mix and workability test has been done and 6 cubes are casted to find the 3Days, 7 days and 28 days strength of the concrete.
- Now, phase I of the project is completed next phase is preparing the concrete by utilizing demolished concrete as a part of coarse aggregate.
- Demolished concrete is collected from the site seeing that it does not contain any chemicals
- Now, using hammer make break the concrete waste into suitable size.
- Physical properties of this demolished concrete are been found and accordingly we adjust the water content for the new mix design and we find strength of the concrete by preparing the cubes of different proportions of coarse aggregate and demolished concrete.
- Now, the cubes are tested for compressive strength and compare for best combination of RCA and CA.

1.3 Thesis Organization

Chapter 1 gives the introduction to the project: the problem identified the objective, proposed outline of the project and scope of the project on dealing with the stated problem.

Chapter 2 gives the summary of the literature review and the final conclusions drawn from them.

Chapter 3 gives the procedures of the tests to be done on different materials used in the preparation of concrete blocks for finding the physical properties which may be useful in finding the mix design.

Chapter 4 gives the description of the mix design; this chapter involves in designing mix ratios for the different combinations of RCA and CA.

Chapter 5 gives the results and their analysis. From these results we have plotted the graphs and analyzed the results

Chapter 6 gives the summary, final conclusions on the course of the project and the future scope of work that can be further carried out

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

The chapter involves the discussion of various research papers reviewed for achieving the aim of the project. The following research papers have discussed about use of recycled concrete in the regular construction activities so that the we can decrease the concrete waste in the environment and the papers have also discussed about the physical properties of the concrete waste.

2.2 Literature review

2.2.1 Recycled coarse aggregates - Sudhir - p patil

- The slump value of the normal concrete is observed to be less than recycled one.
- The compressive strength of concrete containing 50% RCA has strength close to normal concrete.
- Strength is high during initial stages and reduces in later stage.
- Due to lack of treatment process for RCA adequate strength is not achieved.

2.2.2 Experimental Study on Recycled Aggregate Concrete - G. Murali

- In this paper he has considered the treatment of RCA before recycling
- He observed flexural, compressive and split tensile strength of RCA is found to be lower than natural aggregates
- The strength of the RCA was improved by water and acid treatments.
- Among HNo3, H2So4 results of HNo3 were close to natural aggregates.

2.2.3 Characterization of Recycled Aggregate Construction and Demolition Waste for Concrete Production - M. Martin Morals

- In this journal he gives importance to the shape and size of the aggregates.
- The ceramic material and mortar adhering to natural aggregates directly affected physical and mechanical properties of RCA.
- Chemical composition of RCA was weakest as it contains lot of sulphates and chlorides.
- Quality of RCA was improved by blending it with natural aggregates.

2.2.4 Recycled construction and demolition concrete waste as aggregate for structural concrete - Ashraf M. Wagih

- Properties such as absorption and abrasion were lower than the required.
- Fully replacement of natural aggregates by RCA led less workability and decrease in strength.
- This is overcome by using super plasticizers.
- Replacing 25 – 50 % of NCA with RCA god results are achieved.
- Using silica fume as admixture performance of RCA is enhanced as it acts as interfacial zone between old and new mortar.

2.2.5 Use of Recycled Aggregate Concrete - Mr. Tushar R Sonawane

- Use of recycled aggregate up to 30% does not affect the functional requirements of the structure as per the finding of test results.
- Various tests conducted on RAC are found to be satisfactory as per IS 2386.
- Due to the use of recycled aggregate in construction, energy and cost of transportation of NA and excavation is saved.

2.2.6 Strength of Concrete Incorporating Aggregates Recycled from Demolition Waste - R. Kumutha

- Recycled aggregate had influence on the properties of the concrete
- It is possible to use 100% crushed concrete coarse aggregate and 60% crushed brick fine aggregate since target strength is archived
- But there is a considerable decrease in the flexural strength.

2.2.7 Utilization of recycled waste as ingredients in concrete mix- Nitish Puri

- There is a considerable increase in the compressive strength and flexural strength when aggregates are fully or partially replaced with construction debris.
- Maximum strength is achieved having 25% RCA and 75% NA.
- Production cost decreases
- Decrease in slump values is observed

2.2.8 Utilisation of Recycled Aggregate as Coarse Aggregate in Concrete - P.C. Yong

- This is about recycling of tested concrete cubes from RMC's.

- RAC can achieve high compressive strength, split tensile strength as well as flexural strength.
- RAC has higher 28-day compressive strength and higher 28 day split tensile strength compared to control concrete. The 28-day flexural strengths of RAC are lower than natural concrete.

CHAPTER 3
LABORATORY INVESTIGATIONS

3.1 Introduction

The chapter involves in material collection and the laboratory testing of the samples. This involves the collection of samples and various test procedures conducted at the lab. The laboratory testing of the samples is explained in detail along with the procedures and results. These works play an important role in assessing the properties of the samples. The evaluation of the properties by conducting various tests provides the opportunity to determine whether the material is used for construction or not by considering the Indian standard code.

3.2 Fine Aggregate

The material which passes through 4.75 mm sieve and retains on 75 microns sieve such a material is known as fine aggregate this is the important material in the concrete preparation, the strength of the concrete depends up on the type of the fine aggregate used and also quality of the material used if at all low grade material is used then in such case the slump value varies, water absorption varies along with this strength of the material also will also be effected hence always fine aggregate is selected in such a way that all the material properties of such material should satisfy the is code requirements.

3.3 Coarse Aggregate

Material which passes through 4.75mm sieve and retained on 80mm sieve such material is known as coarse aggregate the major part of the concrete is of coarse aggregate out of all the aggregate used in concrete about 60 - 70% of the material is of coarse aggregate hence it is the deciding factor to attain the calculated strength hence the quantity and quality this type of materials is to be taken according to Indian standard only.

3.3.2 Grain Size Analysis

This test is performed to determine the percentage of different grain sizes contained. The mechanical or sieve analysis is performed to determine the distribution of the coarser, larger-sized particles. Grain size analysis provides the grain size distribution, The equipment required are Balance, set of sieves, Cleaning brush.

The stepwise procedure for conducting the grain size analysis is

- Firstly, 1000gm of sample collected
- Prepare a stack of sieves. Sieves having larger opening sizes (i.e., lower numbers) are placed above the ones having smaller opening sizes (i.e., higher numbers). To sieves are arranged in the order of 4.75mm, 2mm, 425microns and 75microns from top to bottom respectively.
- Sieve through a nest of sieves by handshaking. At least 10 minutes of hand sieving is desirable. Remove the stack and carefully weigh and record the weight of each sieve with its retained soil.
- The results of sieve analysis are generally expressed in terms of the percentage of the total weight that passed through different sieves.

3.4 Cement

Cement is the binder material, a substance that sets and hardens and can bind other materials together. The word "cement" traces to the Romans, who used the term opus cementatim to describe masonry resembling modern concrete that was made from crushed rock with burnt lime as binder. The volcanic ash and pulverized brick supplements that were added to the burnt lime, to obtain a hydraulic binder, were later referred to as cementum, cimentum, cament, and cement.

Cements used in construction can be characterized as being either hydraulic or non-hydraulic, depending upon the ability of the cement to be used in the presence of water The most important uses of cement are as a component in the production of mortar in masonry, and of concrete, a combination of cement and an aggregate to form a strong building material

3.5 Recycled Coarse Aggregate

Materials which are dismantled are collected from the respective site and then the physical properties of their materials are found using the below test procedure, if the values of these materials (properties such are as specific gravity, density etc…) are in the range of Indian standards then in such a case these materials are allowed to use in the preparing the cube for finding the compressive strength of the cube.

3.5.2 Grain Size Analysis

This test is performed to determine the percentage of different grain sizes contained. The

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- The results of sieve analysis are generally expressed in terms of the percentage of the total weight that passed through different sieves.

3.6 Compressive Test

Out of many tests applied to the concrete, this is the utmost important which gives an idea about all the characteristics of concrete. By this single test one judge that whether Concreting has been done properly or not. For cube test two types of specimens either cubes of 15 cm X 15 cm X 15 cm or 10cm X 10 cm x 10 cm depending upon the size of aggregate are used. For most of the works cubical moulds of size 15 cm x 15cm x 15 cm are commonly used.

This concrete is poured in the mould and tempered properly so as not to have any voids. After 24 hours these moulds are removed and test specimens are put in water for curing. The top surface of these specimen should be made even and smooth. This is done by putting cement paste and spreading smoothly on whole area of specimen.

These specimens are tested by compression testing machine after 7 days curing or 28 days curing. Load should be applied gradually at the rate of 140 kg/cm2 per minute till the Specimens fails. Load at the failure divided by area of specimen gives the compressive strength of concrete.

APPARATUS

Compression testing machine

PREPARATION OF CUBE SPECIMENS

The proportion and material for making these test specimens are from the same concrete used in the field.

SPECIMEN

6 cubes of 15 cm size Mix. M15 or above

MIXING

Mix the concrete either by hand or in a laboratory batch mixer

HAND MIXING

1. Mix the cement and fine aggregate on a water tight none-absorbent platform until the mixture is thoroughly blended and is of uniform colour
2. Add the coarse aggregate and mix with cement and fine aggregate until the coarse aggregate is uniformly distributed throughout the batch
3. Add water and mix it until the concrete appears to be homogeneous and of the desired consistency

SAMPLING

1. Clean the mounds and apply oil
2. Fill the concrete in the molds in layers approximately 5cm thick
3. Compact each layer with not less than 35strokes per layer using a tamping rod (steel bar 16mm diameter and 60cm long, bullet pointed at lower end)
4. Level the top surface and smoothen it with a trowel

CURING

The test specimens are stored in moist air for 24hours and after this period the specimens are marked and removed from the molds and kept submerged in clear fresh water until taken out prior to test.

PRECAUTIONS

The water for curing should be tested every 7days and the temperature of water must be at 27+-2oC.

PROCEDURE

1. Remove the specimen from water after specified curing time and wipe out excess water from the surface.
2. Take the dimension of the specimen to the nearest 0.2m
3. Clean the bearing surface of the testing machine
4. Place the specimen in the machine in such a manner that the load shall be applied to the opposite sides of the cube cast.
5. Align the specimen centrally on the base plate of the machine.
6. Rotate the movable portion gently by hand so that it touches the top surface of the specimen.
7. Apply the load gradually without shock and continuously at the rate of 140kg/cm2/minute till the specimen fails
8. Record the maximum load and note any unusual features in the type of failure.

Figure 3.1- Finding compressive strength

CHAPTER 4
CONCRETE MIX DESIGN

4.1 Introduction

One of the ultimate aims of studying the various properties of the materials of concrete and hardened concrete is to enable a concrete technologist to design a concrete mix design for a particular strength and durability. The design of concrete mix is not a simple task on account of the widely varying properties of the materials, the conditions that prevail at the site work in particular the exposure condition, and the condition that are demanded for a particular work for which mix is designed. Design of concrete mix design requires complete knowledge of various properties of these constituent materials, the implications in case of change on these conditions at the site, the impact of the properties of plastic concrete on the hardened concrete and the complicated inter-relationships between the variables. All these make the task of mix design more complex and difficult. Design of concrete mix needs not only the knowledge of material properties and properties of concrete in plastic condition, it also needs wider knowledge and experience of concreting. Even then the proportion of the materials of concrete found out at the laboratory requires modification and readjustments to suit the field conditions.

Mix design can be defined as the process of selecting the suitable ingredients of concrete and determining their relative proportions with the object of producing concrete of certain minimum strength and durability as economically as possible. The first objective is to archive the stipulated minimum strength and durability. The second objective is to make the concrete in the most economical manner. Cost wise all concretes primarily depends on two factors namely cost of the material and cost of material. Since the cost of cement is many times more than cost of other ingredients, attention is mainly directed to the use of as little cement as possible consistent with strength and durability.

4.2 Mix Design of Conventional Concrete (M25)

The following procedure is followed to design the concrete

Step 1: Target Mean Strength

$$f'_{ck} = f_{ck} + 1.65 \times \sigma = 25 + 6.6$$

$$= 31.6$$

Table 4.1: Standard Deviation Values

Table 1 Assumed Standard Deviation
(*Clauses* 3.2.1.2, A-3 *and* B-3)

Sl No. (1)	Grade of Concrete (2)	Assumed Standard Deviation N/mm² (3)
i)	M 10	3.5
ii)	M 15	
iii)	M 20	4.0
iv)	M 25	
v)	M 30	5.0
vi)	M 35	
vii)	M 40	
viii)	M 45	
ix)	M 50	
x)	M 55	

NOTE — The above values correspond to the site control having proper storage of cement; weigh batching of all materials; controlled addition of water; regular checking of all materials, aggregate grading and moisture content; and periodical checking of workability and strength. Where there is deviation from the above, values given in the above table shall be increased by 1 N/mm².

Step 2: W/C Ratio

- From the graph specified by IS10262, w/c is taken as 0.46 for 25 N/mm^2 compressive strength of concrete.
- From the Table 5 of Is 456, for moderate condition, w/c ratio is 0.45
- Hence, min of two values i.e., 0.45 is taken as w/c ratio.

Step 3: Water Content

From Table 2 of IS10262, assume 20mm of aggregate is been used and hence max water content is 186 kg's, slum is assumed to be 100mm and hence water should be increases by 6%

$$\text{Water content} = 186 \times 1.06 = 197.16$$

Table 4.2: Water Content

Table 2 Maximum Water Content per Cubic Metre of Concrete for Nominal Maximum Size of Aggregate
(*Clauses* 4.2, A-5 *and* B-5)

Sl No.	Nominal Maximum Size of Aggregate mm	Maximum Water Content [1] kg
(1)	(2)	(3)
i)	10	208
ii)	20	186
iii)	40	165

NOTE — These quantities of mixing water are for use in computing cementitious material contents for trial batches.

[1] Water}content corresponding to saturated surface dry aggregate.

Table 4.3: Minimum Water Content, Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures

CONCRETE MIX DESIGN

Table - 5

Minimum Cement Content, Maximum Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregate of 20 mm Nominal Maximum Size

IS:456-2000

Sl No.	Exposure	Plain Concrete			Reinforced Concrete		
		Minimum Cement Content kg/m^2	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete	Minimum Cement Content kg/m^2	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete
1	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Mild	220	0.60	-	300	0.55	M20
ii)	Moderate	240	0.60	M15	300	0.50	M25
iii)	Severe	250	0.50	M20	320	0.45	M30
iv)	Very Severe	260	0.45	M20	340	0.45	M35
v)	Extreme	280	0.40	M25	360	0.40	M40

Notes

1 Cement content prescribed in this table is irrespective of the grades of cement and it is inclusive of additions mentioned in 5.2 additions such as fly ash or ground granulated blast furnace slag may be taken into account in the concrete composition with respect to the cement content and water-cement ratio if the suitability is established and as long as the maximum amounts taken into account do not exceed the limit of Pozzolana and slag specified in IS: 1489 (Part-1) and IS: 455 respectively.

Step 4: Calculation of Cement Content

$$\frac{w}{c} = 0.45$$

hence,

$$c = \frac{197.16}{0.45} = 438.13$$

from table of IS456,

min cement content = 320 and hence we use 438.13 Kg/m^3

Step 5: Calculation of Coarse Aggregate and Fine Aggregate

From IS10262 we have classified the zone of fine aggregate as Zone I and corresponding volume of coarse aggregate for 20mm size is 0.62 if w/c is 0.5 but our w/c is 0.46 hence the effective volume of coarse aggregate is 0.6, for w/c 0.42 volume of coarse aggregate is found as 0.608

and hence volume of fine aggregate is 1 - 0.608 = 0.392

Table 4.4: Zone classification

Table 3 Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate
(*Clauses* 4.4, A-7 *and* B-7)

Sl No.	Nominal Maximum Size of Aggregate mm	Volume of Coarse Aggregate[1] per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate			
		Zone IV	Zone III	Zone II	Zone I
(1)	(2)	(3)	(4)	(5)	(6)
i)	10	0.50	0.48	0.46	0.44
ii)	20	0.66	0.64	0.62	0.60
iii)	40	0.75	0.73	0.71	0.69

[1] Volumes are based on aggregates in saturated surface dry condition.

Step 6: Mix Proportion

volume of concrete = 1m^3

$$\text{volume of cement} = \frac{438.13}{(3.2 \times 1000)}$$

= 0.136m^3

volume of water = 197.16

= 0.197m^3

Absolute weight of all materials except total aggregates = 1 - (0.136 + 0.197)

= 0.667

volume of coarse aggregate = 0.667×0.61×2.68×1000

= 1090.41 m^3

volume of fine aggregate = 0.667×0.39×2.52×1000

= 655.52 m^3

Step 7: Mix Ratio

438.13: 655.52: 1090.41

1:1.49:2.48

4.3 Mix Design of Recycled Concrete

The following procedure is followed to design the concrete

Step 1: Target Mean Strength

$$f_{ck}' = f_{ck} + 1.65 \times \sigma$$

= 25 + 6.6

=31.6

Step 2: $\frac{W}{C}$ Ratio

- From the graph specified by IS10262, is taken as 0.46 for 25 N/mm^2 compressive strength of concrete.
- From the Table 5 of Is 456, for moderate condition, w/c ratio is 0.45
- Hence, min of two values i.e., 0.45 is taken as w/c ratio.

Step 3: Water Content

From Table 2 of IS10262, assume 20mm of aggregate is been used and hence max water content is 186 kg's, slum is assumed to be 100mm and hence water should be increases by 6%

Water content = $186 \times 1.06 = 197.16$

Step 4: Calculation of Cement Content

$\frac{W}{C}$ =0.45 hence c= 197.16/0.45 = 438.13

from table of IS456, min cement content = 320 and hence we use 438.13 Kg/m^3

Step 5: Calculation of Coarse Aggregate and Fine Aggregate

From IS10262 we have classified the zone of fine aggregate as Zone I and corresponding volume of coarse aggregate for 20mm size is 0.62 if w/c is 0.5 but our w/c is 0.46 hence the effective volume of coarse aggregate is 0.6, for w/c 0.42 volume of coarse aggregate is found as 0.608

and hence volume of fine aggregate is 1 - 0.608 = 0.392

Step 6: Mix Proportion

volume of concrete = 1m^3

volume of cement = $\frac{438.13}{3.2 \times 1000}$

= 0.136m^3

volume of water = 197.16

= 0.197m^3

Absolute volume total aggregates = 1 - (0.136 + 0.197)

= 0.667m^3

weight of coarse aggregate = 0.667×0.61×2.22×1000

= 903.25

weight of fine aggregate = 0.667×0.39×2.52×1000

= 655.52

Step 7: Mix Ratio

438.13: 655.52: 903.25

1:1.49:2.06

4.4 Mix Design for Different Combinations of Natural and Recycled Aggregates

All the steps above will be same in all the cases only the ratio of changes as the effective specific gravity changes i.e., only last two steps will change in each case

4.4.1 Mix Ratio for Combination Of 50% CA And 50% RCA

Step 1: Effective Specific Gravity

$G' = (0.5\times2.68) + (0.5\times2.22) = 2.45$

Step 2: Mix Proportion

volume of concrete = $1m^3$

volume of cement = $\frac{438.13}{3.2\times1000}$

$= 0.136m^3$

volume of water = 197.16

$= 0.197m^3$

Absolute volume total aggregates = 1 - (0.136 + 0.197)

$= 0.667m^3$

weight of coarse aggregate = $0.667\times0.61\times2.45\times1000$

= 996.83

weight of fine aggregate = $0.667\times0.39\times2.52\times1000$

= 655.52

Step 3: Mix Ratio

438.13: 655.52: 996.83

1:1.49:2.27

4.4.2 Mix Ratio for Combination Of 25% CA And 75% RCA

Step 1: Effective Specific Gravity

$G = (0.25\times2.68) + (0.75\times2.22)$

$=2.335$

Step 2: Mix Proportion

volume of concrete = $1m^3$

volume of cement = $\frac{438.13}{3.2\times1000}$

$= 0.136m^3$

volume of water = 197.16

= $0.197m^3$

Absolute volume total aggregates = 1 - (0.136 + 0.197)

= $0.667m^3$

weight of coarse aggregate = 0.667×0.61×2.335×1000

= 950.04

weight of fine aggregate = 0.667×0.39×2.52×1000

= 655.52

Step 3: Mix Ratio

438.13: 655.52: 950.04

1:1.49:2.16

4.4.3 Mix Ratio for Combination Of 75% CA And 25% RCA

Step 1: Effective Specific Gravity

G= (0.75×2.68) + (0.25×2.22)

=2.565

Step 2: Mix Proportion

volume of concrete = $1m^3$

volume of cement = $\frac{438.13}{3.2 \times 1000}$

= $0.136m^3$

volume of water = 197.16

= $0.197m^3$

Absolute volume total aggregates = 1 - (0.136 + 0.197)

= $0.667m^3$

weight of coarse aggregate = 0.667×0.61×2.565×1000

= 1043.62

weight of fine aggregate = 0.667×0.39×2.52×1000

= 655.52

Step 3: Mix Ratio

438.13: 655.52: 1043.62

1:1.49:2.38

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter deals with the compressive test results, cubes of different mixes which are discussed earlier are prepared, for each mix design 6 cubes are made so that tests on compression can be done on two cubes for 7 days,14 days and 28 days and average value is taken as compressive strength, average value is considered generally it is because to minimize the errors.

5.3 Compressive Test Results

Table 5.2: Compressive Strength

Type of concrete	7 days	14 days	21 days
CA	**18.82**	**22.4**	**27.9**
RCA	**17.84**	**19.04**	**25.77**
50%RCA+50%CA	**16.29**	**20.01**	**24.06**
25%RCA+75%CA	**18.14**	**19.54**	**-**
75%RCA+25%CA	**14.05**	**15.6**	**-**

5.4 Comparative Graphs

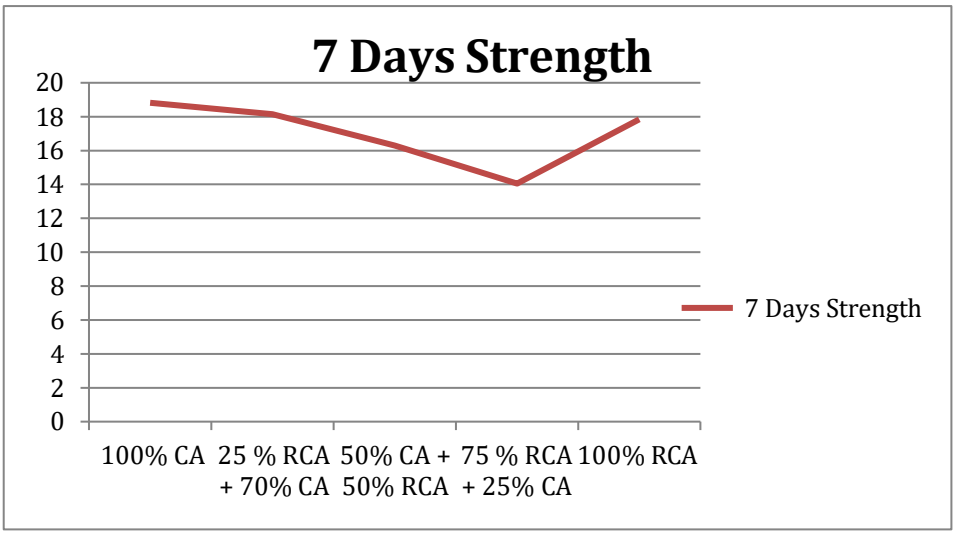

Figure 5.1 – 7 Days strength of different proportions of concrete

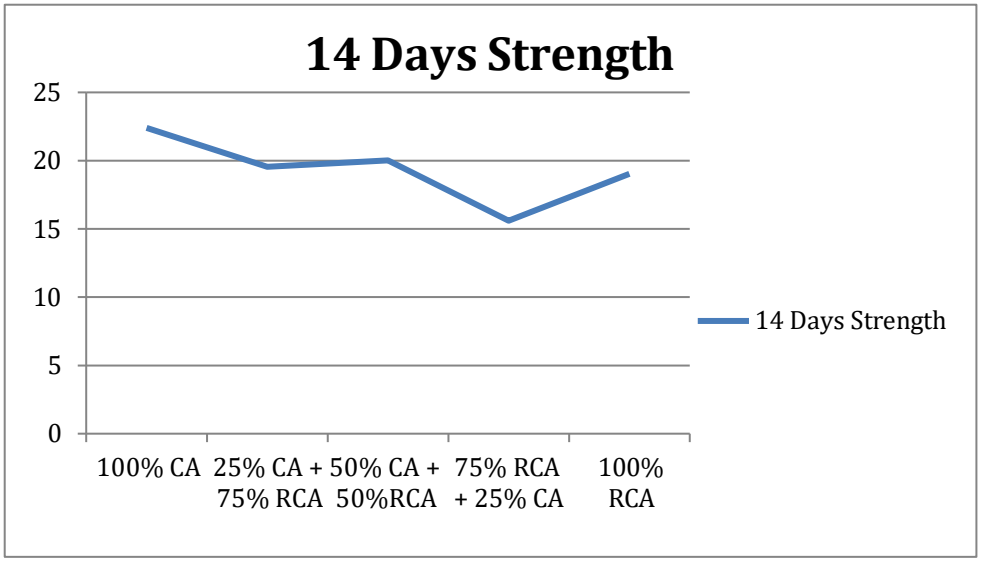

Figure 5.2 – 14 Days strength of different proportions of concrete

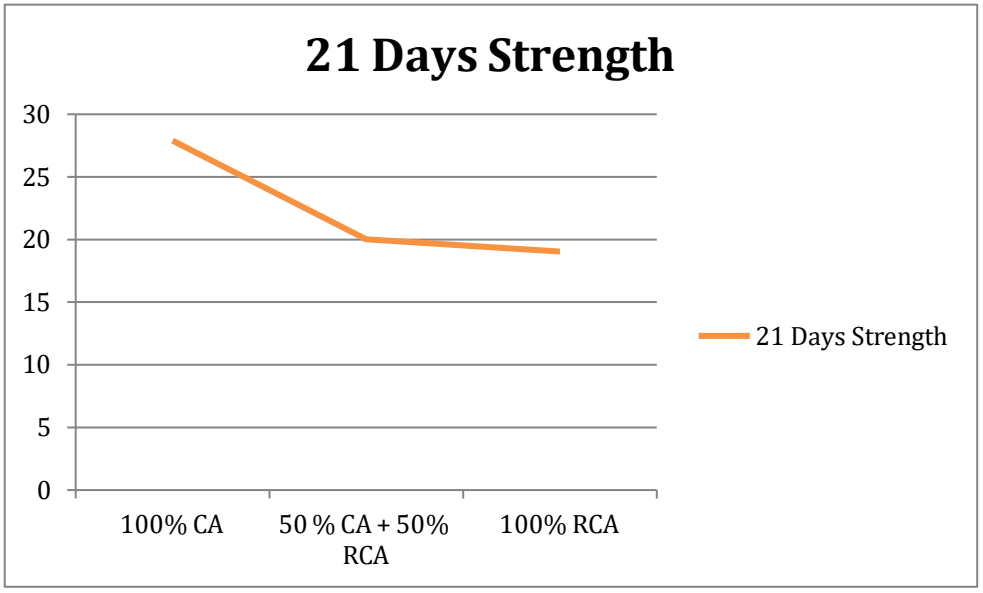

Figure 3 – 21 Days strength of different proportions of concrete

CHAPTER 6
CONCLUSION

6.1 Summary

Concrete is a combination of Cement, Natural sand, and Aggregates. Cement is a binding property and a mixture of several chemical compounds. Aggregates are granites which are indigenous rocks which are excessively available. Natural sand is river sand which is declining day by day and its utilization is increasing day by day. Because of this reason it is very important to find an alternative for this problem.

Dismantled RCC structure produce heavy concrete waste is a waste. This is abundantly available and usually dumped in the earth's crust, due to which fertility of the soil decrease and land becomes useless, to avoid this kind of problems concrete waste can be reused in the concrete and can be used in construction.

Concrete waste can be partially replaced as coarse aggregate in construction. Based upon the requirement we can change the % of CA replaced by RCA so that the construction becomes economical, we can also use 100% RCA as CA for concrete walls used as partisan walls as there will be no loads on the structure.

Demolished aggregate is collected from site and then this aggregate is broken into 20mm size coarse aggregate and then different parameters have to be evaluated such as specific gravity, sieve analysis for natural sand as. Other parameters like water absorption test, fineness of cement is evaluated for aggregates and cement. These all parameters are required in order to obtain mix design of concrete moulds. When mix design is evaluated as per IS code, concrete moulds are prepared and compressive strength of them is known for every 3, 7, and 21days, tensile strength is found for 21 days.

With every 100%, 75%, 50%, and 25% replacement of RCA with CA, compressive strength has to be evolved, and comparative studies have to be made between conventional concrete and partially replaced concrete.

6.2 Conclusion

1. Various tests conducted on RCA are compared with Indian code and the results are satisfactory and hence these can be used as aggregates
2. Due to use of RCA in construction energy, cost of transportation is saved.

3. Up to 50% replacing of RCA we get satisfactory strength
4. Production cost decreases remarkably
5. Due to lack of treatment of RCA adequate strength is not archived but by applying some treatment processes we can further improve the strength of the RCA
6. Tensile test shows concrete has good tensile strength when replaced 50%
7. Water absorption of RCA is high when compared with conventional aggregate.

6.3 Future Scope

Further we can still investigate whether the increase in the strength of concrete is decreasing or increasing because as we are using RCA due to presents of some chemical impurities there may be a chance of decrease in the strength suddenly so by further investigation, we can find the faults and hence we can find out some treatment for that in the initial stage and hence this type of mix can be feasible in the future.